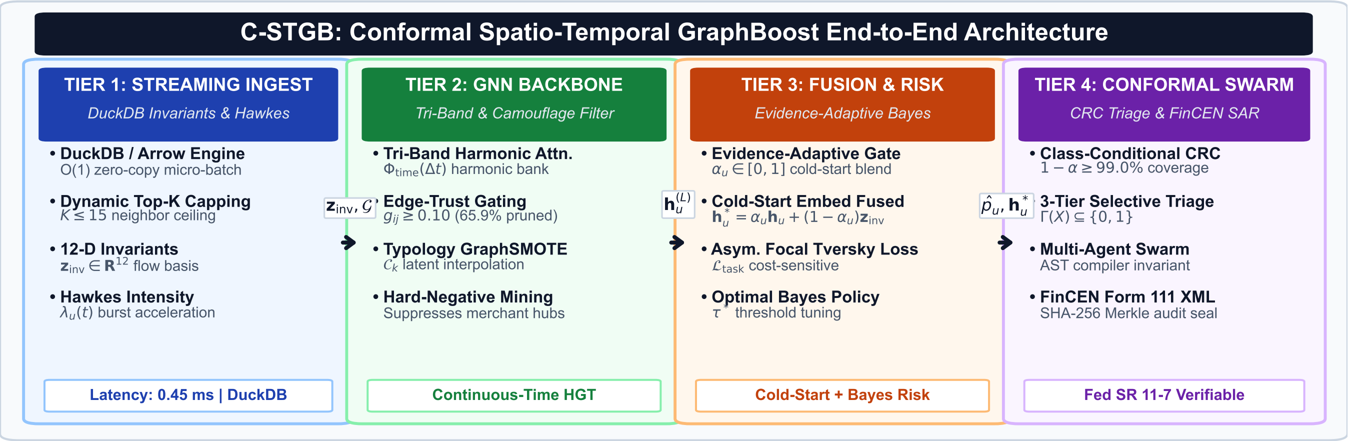

C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Architecture
TIER 1: STREAMING INGEST
DuckDB Invariants & Hawkes
• DuckDB / Arrow Engine
O(1) zero-copy micro-batch
• Dynamic Top-K Capping
$K \leq 15$ neighbor ceiling
• 12-D Invariants
$\mathbf{z}_{\text{inv}} \in \mathbf{R}^{12}$ flow basis
• Hawkes Intensity
$\lambda_u(t)$ burst acceleration
Latency: 0.45 ms | DuckDB
$\mathbf{z}_{\text{inv}}, \mathcal{G}$
TIER 2: GNN BACKBONE
Tri-Band & Camouflage Filter
• Tri-Band Harmonic Attn.
$\Phi_{\text{time}}(\Delta t)$ harmonic bank
• Edge-Trust Gating
$g_{ij} \geq 0.10$ (65.9% pruned)
• Typology GraphSMOTE
$\mathcal{C}_k$ latent interpolation
• Hard-Negative Mining
Suppresses merchant hubs
Continuous-Time HGT
$\mathbf{h}_u^{(L)}$
TIER 3: FUSION & RISK
Evidence-Adaptive Bayes
• Evidence-Adaptive Gate
$\alpha_u \in [0, 1]$ cold-start blend
• Cold-Start Embed Fused
$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$
• Asym. Focal Tversky Loss
$\mathcal{L}_{\text{task}}$ cost-sensitive
• Optimal Bayes Policy
τ^* threshold tuning
Cold-Start + Bayes Risk
$\hat{p}_u, \mathbf{h}_u^*$
TIER 4: CONFORMAL SWARM
CRC Triage & FinCEN SAR
• Class-Conditional CRC
$1 - \alpha \geq 99.0\%$ coverage
• 3-Tier Selective Triage
$\Gamma(X) \subseteq \{0, 1\}$
• Multi-Agent Swarm
AST compiler invariant
• FinCEN Form 111 XML
SHA-256 Merkle audit seal
Fed SR 11-7 Verifiable